

## EDUCATION

**Worcester Polytechnic Institute (WPI)**  
*Doctorate of Philosophy, Robotics Engineering*  
Total GPA via first 36 credits: **4/4**

Aug 2024 – Present  
Worcester, MA

**Iran University of Science and Technology (IUST)**  
*Bachelor of Science, Mechanical Engineering*  
Total GPA: **16.12/20 (3.34/4)**

Sep 2018 – Sep 2023  
Tehran, Iran

**National Organization for Development of Exceptional Talents (NODET)**  
*High School Diploma in Mathematics and Physics*  
Total GPA: **19.01/20 (4/4)**

Sep 2014 – Jul 2018  
Tehran, Iran

## HONORS AND AWARDS

- **Best Student Paper Award** at IEEE International Conference on Robotics and Automation (ICRA) 2026.
- **Best Student Paper Award** at Hellenic Robotics Forum (HRF) 2026.
- **Glenn Yee 2026 Travel Award** of Robotics Engineering Graduate Student.
- **Top 10%** within Mechanical Engineering students class of 2022.
- **Top 0.75%** among 144437 students in the National University Entrance Exam 2018.
- **Ranked 4<sup>th</sup>** at Robocup 2014 Iran Open International Competitions (Soccer 2D Simulation).

## PUBLICATIONS

Zhong, Z., Golestaneh, S., Chamzas, C., **ActivePusher: Active Learning and Planning with Residual Physics for Nonprehensile Manipulation**, *ICRA 2026* [Best Student Paper at ICRA 2026 and HRF 2026] [Link](#)

Golestaneh, S., Zhong, Z., Lee, D., Chamzas, C., **AURA: Asymptotically Optimal Uncertainty-Robust Replanning Algorithm for Kinodynamic Systems** [IEEE Robotics and Automation Letters] [Link](#)

Yazdipaz, K., Kohli, N., Golestaneh, S.A., Shahbazi, M., **Robust and Efficient Phase Estimation in Legged Robots via Signal Imaging and Deep Neural Networks**, *IEEE Access* [Link](#)

Zhong, Z., Golestaneh, S., Chamzas, C., **Terminal Matters: Kinodynamic Planning with a Terminal Cost and Learned Uncertainty in Belief State-Cost Space** [Under Review at IEEE Transaction on Robotics] [Link](#)

Golestaneh, S.\*, Lee, D.\*, Singh, J., Zhong, Z., Chamzas, C., **MetaPusher: Meta Learning and Planning for Nonprehensile Manipulation of Unseen Objects with Rapid Online Adaption** [Under Review at IEEE International Conference on Robotics and Automation] [Link](#)

## SKILLS

**Programming Languages:** C++, Python, Matlab, EES

**Engineering Software :** Genesis, MuJoCo, SolidWorks, CATIA, ADAMS, Arduino

**Operating Systems:** Linux, ROS, Windows

**Publishing Tools:**  $\text{\LaTeX}$ , Microsoft Office Package, Vim

**Languages:** Persian (Native), English (Fluent)

- **IELTS: 7 → L: 8.5 R: 6.5 W: 6 S: 6.5**
- **GRE: 312 → V:142 Q:170**

## PROJECTS

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### Ph.D Directed Research

Efficient Motion Planning for Kinodynamic Systems and Nonprehensile Manipulation

Aug 2024 – Present  
Worcester, MA

- Extensive review of geometric and kinodynamic motion planners.
- Enhancing the efficiency of motion planning algorithms to maximize the accuracy.
- Implementing UR10/UR16e robot trajectory control methods in different simulation environments.
- Training Physics Residual Neural Networks to boost dynamics model efficiency with limited data.

### Humanoid Robotics Development Project

Unitree G1 Development and Control

Jan 2026  
Worcester, MA

- Developing and testing locomotion gait modes for the Unitree G1 humanoid robot.
- Implementing whole body motion control for stability.
- Integrating robot control commands within the Unitree G1 software framework and physical robot.

### Robotics Control Project

Control and Plan for OpenManipulatorX

Nov 2024  
Worcester, MA

- Implementing ROS-based forward and inverse kinematics of the robot arm.
- Applying kinematic velocity control through Jacobian inversion.
- Performing optimization of PD and torque controllers through gain calibration.

### Bachelor's Dissertation

Analysis and Implementation of a Quadruped Robot Simulator

Oct 2021 – Sep 2023  
Tehran, Iran

- Comprehensive study of legged robot gait algorithms and robotic arm dynamics.
- Building and testing MIT Mini Cheetah simulation and MiLAB robot simulation using Qt interface and ROS.
- Developing IUST quadruped robot simulator based on similar simulation environments.
- Adding a new joint controller for debugging the initial quadruped robot tests.

### Mechanical Engineering Internship

Rahe Andisheh Company

Jul 2020 – Sep 2020  
Tehran, Iran

- Developing a shrimp pond oxygen machine using reverse engineering.
- Redesigning the chassis design using SolidWorks to meet the company's new needs.
- Gearbox and actuator selection based on machine weight distribution.
- Checking structural features along with manufacturing process and testing.

## WORK EXPERIENCE

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### Research Assistant

- *Learning Dynamics for Manipulation and Motion Planning*  
Supervisor: Dr. C. Chamzas, ELPIS Laboratory at WPI
- *Simulation and Construction of Legged Robots*  
Supervisor: Dr. M. Shahbazi, Mechatronics Laboratory at IUST

Aug 2024 – Present  
Worcester, MA  
Sep 2021 – Jul 2024  
Tehran, Iran

### Teaching Assistant

- *Motion Planning and Machine Learning for Robotics*  
Supervisors: Dr. C. Chamzas
- *Mechanical Applications in Robotics and Sensing and Perception in Robotics*  
Supervisors: Dr. A. Rosendo and Dr. G. Lewin
- *Fundamentals of Computer Programming*  
Supervisor: Dr. M. Shahbazi
- *Vectorial Dynamics and Dynamics of Machinery*  
Supervisors: Dr. S. Khanmirza and Dr. S. Asiaei

Aug 2025 – May 2026  
Worcester, MA  
Jan 2025 – May 2025  
Worcester, MA  
Sep 2023 – Feb 2024  
Tehran, Iran  
Sep 2022 – Feb 2023  
Tehran, Iran

### High School Teacher

*Physics Teacher at National Organization for Development of Exceptional Talents High School*

Sep 2021 – May 2024  
Tehran, Iran